

PROFIT LEAKAGE & DISCOUNT OPTIMIZATION ANALYSIS



Dataset:

Superstore Sales Dataset

TOOLS USED

- PYTHON
- PANDAS
- MATPLOTLIB
- SEABORN
- GOOGLE COLAB

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BUSINESS PROBLEM

The company uses discounts to increase sales performance; however, excessive discounting may reduce profitability.

This case study aims to identify:

- areas of profit leakage
- impact of discounts on profitability
- high-risk sales categories
- loss-making products
- regional business performance

Objective:

To provide business recommendations that can improve profitability and reduce risky sales practices.

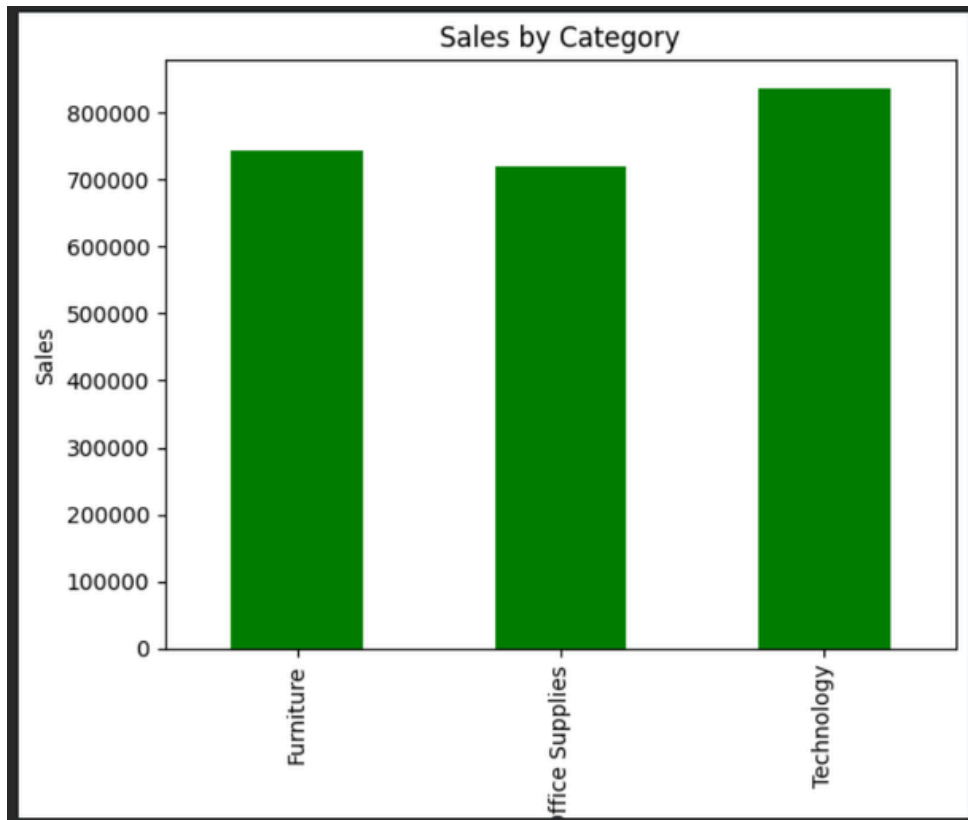
Dataset Overview

<u>Metric</u>	<u>Value</u>
Total Rows	9994
Total Columns	21
Categories	3
Region	s 4

Important Variables

<u>Variable</u>	<u>Meaning</u>
Sales	Revenue generated
Profit	Profit earned
Discount	Discount provided
Category	Product type
Region	Geographic region

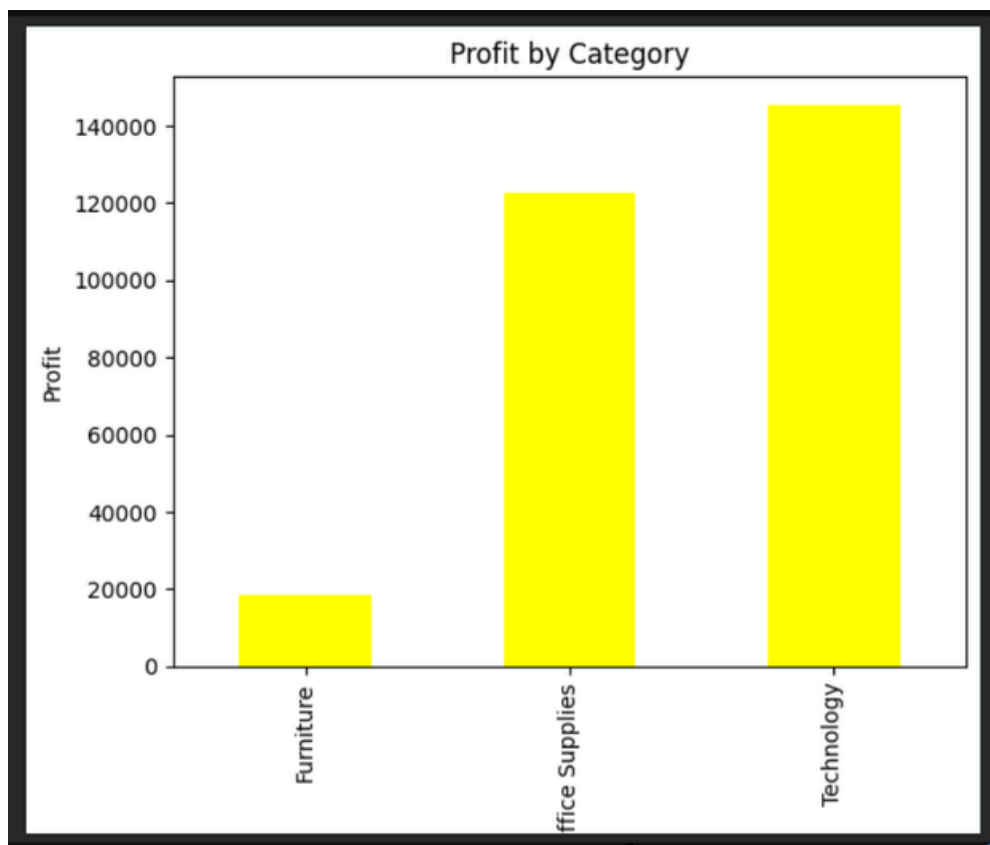
SALES BY CATEGORY



Insight

Technology category generated the highest sales volume, showing strong customer demand.

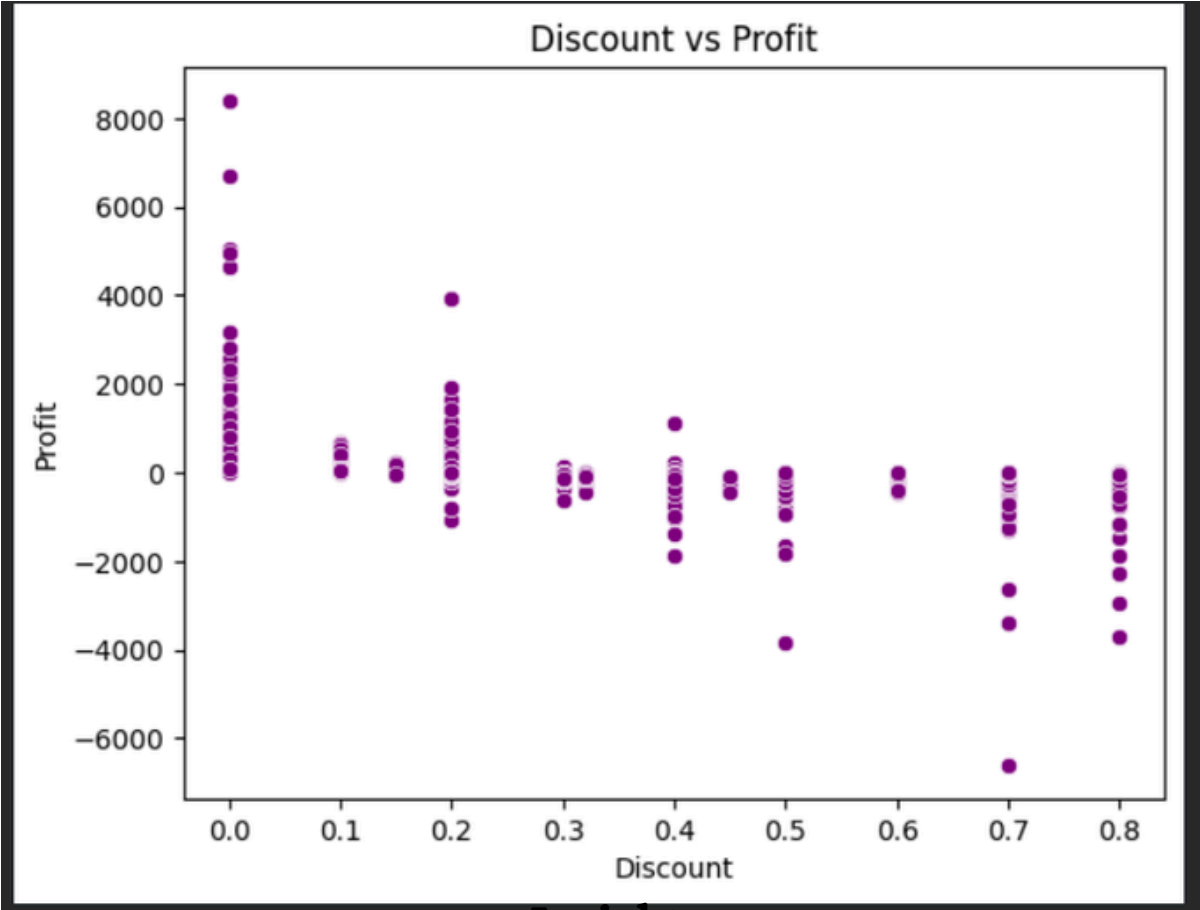
PROFIT BY CATEGORY



Insight

Although sales were high across categories, profitability varied, indicating operational differences and pricing impact.

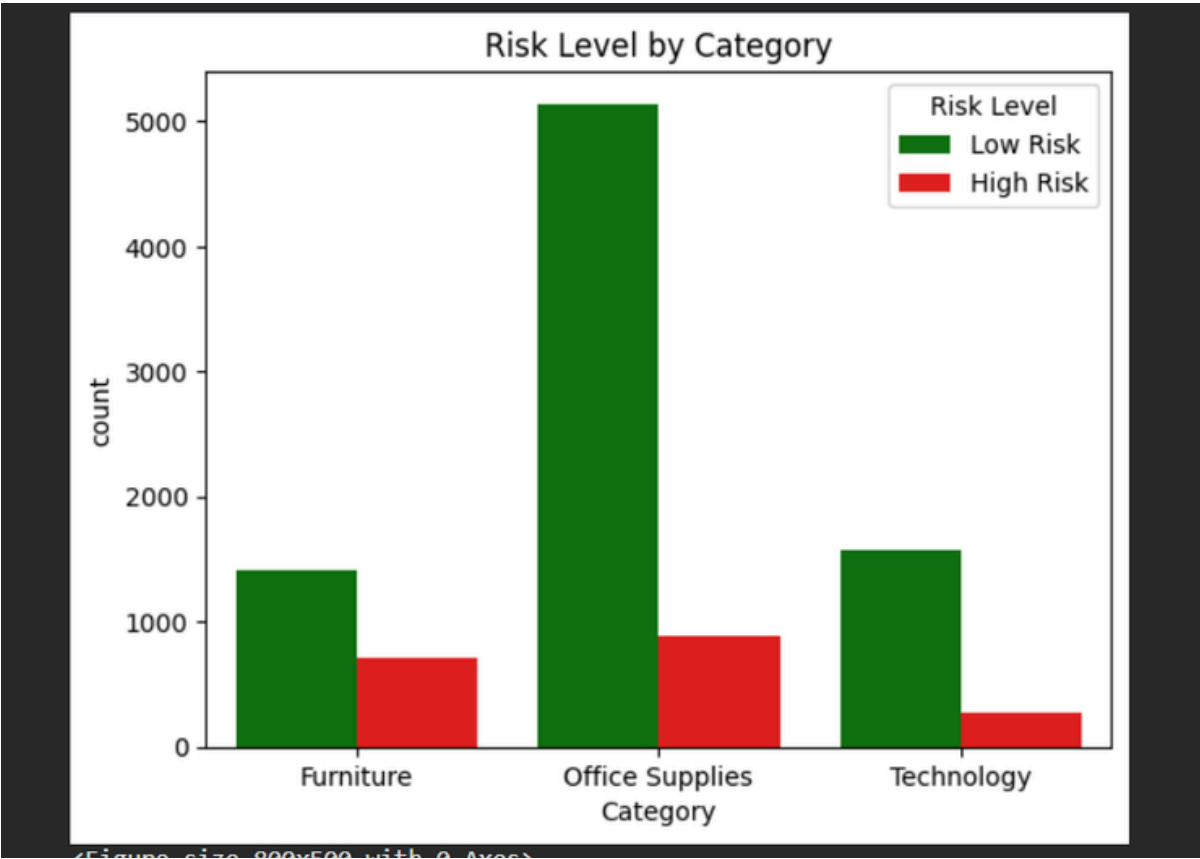
DISCOUNT VS PROFIT



Insight

Higher discount values are frequently associated with lower profitability, indicating that aggressive discounting contributes to profit leakage.

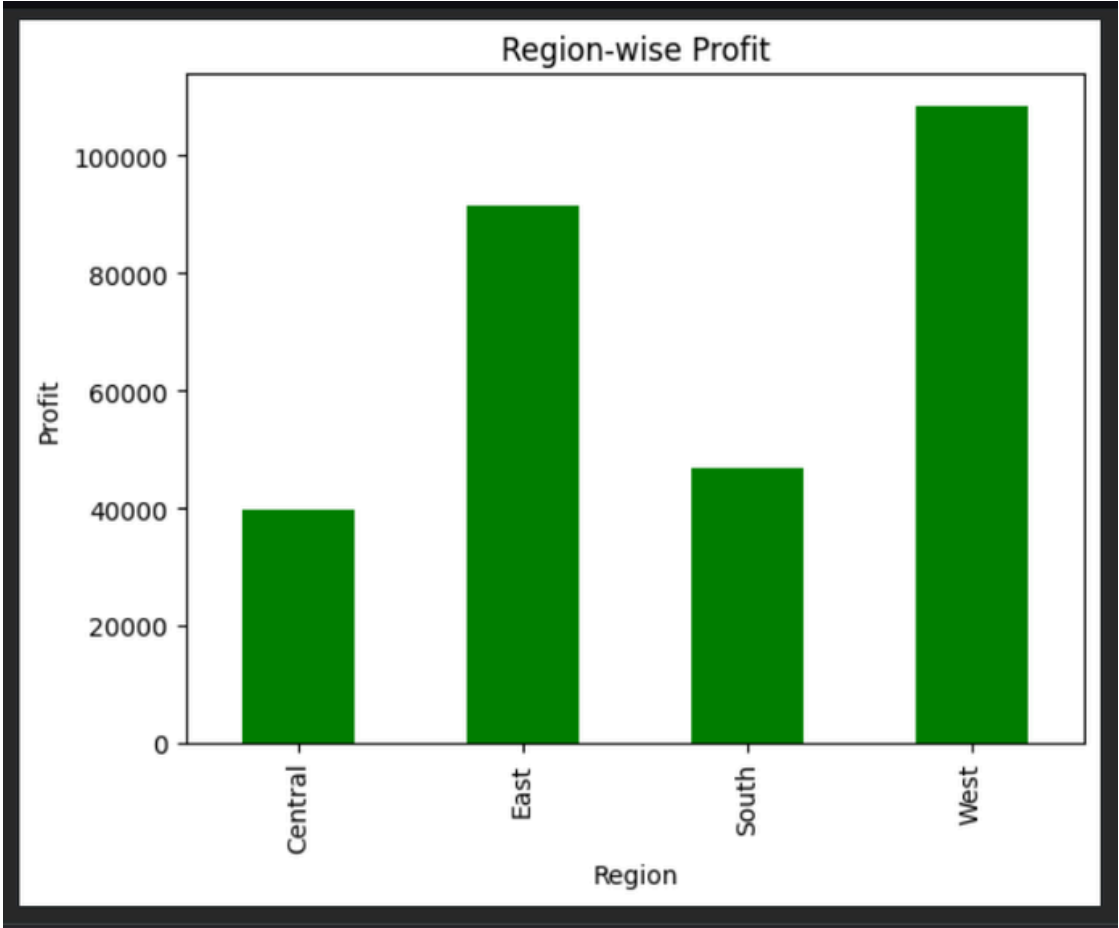
RISK LEVEL BY CATEGORY



Insight

Furniture showed a relatively higher number of high-risk transactions compared with Technology, indicating pricing or discount inefficiencies.

REGION-WISE PROFIT



Insight

Regional profitability differed significantly, suggesting location-based operational performance differences.

TOP LOSS-MAKING PRODUCTS



Insight

Several products generated repeated losses despite sales activity, indicating a need for pricing or inventory review.

RECOMMENDATIONS

- Reduce excessive discounts above a threshold to minimize profit leakage.
- Monitor high-risk categories for pricing inefficiencies.
- Reassess loss-making products to improve profitability.
- Focus on regions showing stronger profit performance.
- Develop better discount approval strategies.

KEY FINDINGS

- High discounts reduce profitability
- Furniture category contains higher business risk
- Several products consistently generate losses
- Regional profitability varies significantly

CONCLUSION

This case study identified several drivers of profit leakage, including excessive discounting and loss-making products.

Using Python-based analysis and visualization, the study highlighted:

- discount impact on profit
- category-level risk patterns
- regional profitability differences
- high-risk business areas

The findings can support better pricing, discount, and profitability decisions.